Review round: 1

Reviewer: 1

The manuscript presents an integrated soft wearable platform for cardiopulmonary acoustic sensing, real-time respiratory sound classification, and vibrotactile feedback. The system-level integration is technically attractive, and the device-level demonstrations are promising. The authors should address below comments for further consideration.

1. Several claims are currently overstated. The system mainly classifies respiratory sound patterns, such as wheeze, crackle, and rhonchi, rather than diagnosing respiratory diseases. In addition, the vibrotactile module demonstrates recognizable feedback patterns, but does not yet show clear behavioral or clinical benefit.

2. The estimated heart rate and breathing rate should be validated against gold-standard or clinically approved reference devices.

3. The manuscript should report the power consumption of the full sensing–actuation system and the expected operating time with a fully charged battery.

4. The current ML model classifies short 2-s windows into sound classes such as wheeze, crackle, rhonchi, breathing, and non-breathing. If every abnormal window triggers feedback, the system may generate excessive alerts and cause user habituation. The authors should aggregate predictions over multiple breaths or longer respiratory episodes, define clinically meaningful alert criteria, and clarify what corrective action each tactile cue is intended to guide.

5. The benefit of reproducing a user's normal breathing profile as tactile feedback is unclear. If the goal is respiratory pacing, simpler cues such as sinusoidal, ramped, or box-breathing patterns may be sufficient. The authors should show that the personalized breath-mimicking profile improves entrainment, breathing regularity, comfort, or physiological outcomes compared with simpler matched cues.

6. Fig. 5E–H shows that the system can extract S1/S2 timing and reproduce a heartbeat-like tactile pattern. Since heartbeat timing is not directly actionable by the user, what is the clinical meaning for giving a heartbeat-like tactile cue. The authors should either provide behavioral and physiological evidence of stress reduction or reframe this section as a technical demonstration of personalized tactile pattern generation.

7. The rationale for the selected tactile cues remains unclear. The manuscript shows that different respiratory sounds can be mapped to distinct vibration patterns, but it does not demonstrate that these cues are clinically actionable or that following them improves breathing. The authors should clarify what response each cue is intended to elicit and provide evidence of benefit, such as reduced wheeze burden, improved breathing regularity, oxygen saturation, or clinical respiratory score.

Reviewer: 2

This paper presents a development of soft wearable device listens to breathing, recognizes warning signs such as wheezing, and provides personalized vibrations to guide breathing, helping people manage respiratory diseases earlier and more safely.

This study is expected to provide a useful wearable healthcare device and this can be strongly recommended for publication with the following revision suggestions,

1) The classification system identifies healthy breathing, wheeze, crackle, and rhonchus. This is useful, but the manuscript should explain more clearly what each sound means in everyday clinical practice. These sounds are not diseases by themselves, and the same sound may occur in several different respiratory conditions. Wheezing may be associated with asthma, airway narrowing, or other causes, while crackles may have several possible clinical explanations. The text should therefore avoid suggesting that the device directly diagnoses a specific disease. A clearer distinction between sound recognition and medical diagnosis would make the clinical claims more accurate.

2) The manuscript suggests that the system may enable earlier intervention and prevent respiratory deterioration. This is an important possibility, but the current experiments mainly demonstrate signal recording, sound classification, and delivery of tactile cues. They do not establish that the device prevents exacerbations, improves treatment adherence, reduces hospital visits, or improves patient outcomes. The discussion should clearly separate demonstrated findings from future clinical possibilities. Statements about prevention should be presented as potential applications rather than established benefits.

3) The manuscript states that pulmonologists provided the respiratory sound labels and clinical diagnoses were used as ground truth. However, the exact labeling process requires greater explanation. It is unclear whether each recording was reviewed by one physician or several physicians, whether the reviewers worked independently, and how disagreements were resolved. The manuscript should specify whether labels represented the dominant sound, the presence of any sound, or a clinical diagnosis assigned to the participant.

4) The study includes 43 participants, consisting of healthy adults, adult patients, and children with persistent asthma. This combination is valuable, but the manuscript should describe the participant population more fully. Important information includes age ranges, sex distribution, disease severity, medication status, smoking history, and the specific respiratory conditions represented among the adult participants.

5) The manuscript reports the use of group based data splitting to avoid sharing recordings from the same participant across data sets. This is a positive methodological choice. However, the exact number of participants and recordings in each data set should be stated clearly.

6) The reported classification accuracy of at least 85 percent across respiratory classes is encouraging. Nevertheless, accuracy alone does not show whether the system performs equally well for every class or whether false alarms are common. The manuscript should provide sensitivity, specificity, precision, recall, and the F1 score for each respiratory sound category.

7) The manuscript presents a meaningful and promising wearable platform that combines respiratory sound monitoring, machine learning classification, and immediate tactile feedback. Recent progresses in the respiratory monitoring/protection systems (Biomaterials, 314, 122866 (2024); ACS Nano, 15, 15730 (2021); Adv. Mater., 36, 2405568, (2024)) should be briefly discussed in the introduction part to provide the readers with the recent related progresses.

8) The paper describes the system as operating in real time, using two second windows updated every 0.5 seconds. This description is understandable, but the practical delay between sound acquisition, signal processing, classification, wireless transmission, and vibration delivery should be reported explicitly. The authors should state the average and maximum processing latency and explain whether delays differ between classification categories.

9) The use of two second windows with a 0.5 second update interval provides a simple framework for continuous classification. However, fixed windows may divide a respiratory event between two segments or miss short events that do not occupy enough of the window. Padding shorter segments may also introduce artificial patterns. The manuscript should discuss how these choices could influence detection of brief crackles, changing wheeze patterns, or irregular breathing.

10) The vibration patterns are designed to represent different respiratory sound classes, and participants could distinguish them with high accuracy. However, recognizing a vibration pattern is not equivalent to understanding what action should follow. The manuscript should explain how a patient is expected to respond to each tactile alert.

Reviewer: 3

Summary

This manuscript proposes a wearable device that integrates real-time sensing, diagnosis, and feedback of cardiopulmonary signals through acoustic sensing and vibrotactile actuation. The machine learning-based diagnostic approach demonstrates high accuracy, and the concept of actively intervening with patients based on diagnostic outcomes marks a meaningful advancement toward interactive medical devices.

However, the personalized feedback mechanism—arguably the most compelling aspect of this work—would benefit from a clearer articulation of its intended therapeutic objective beyond functioning as an alert system. Additional evidence supporting the therapeutic value of this feedback, or a more carefully scoped discussion of its intended role, would further enhance the impact of the study.

Overall, I believe the proposed system is novel and technically strong. Addressing the points above—particularly by clarifying the intended therapeutic role of the feedback system and appropriately supporting or moderating the corresponding claims—would substantially strengthen the manuscript and make it suitable for publication.

Major Comments

1. Measurement location: It is not clear whether the reported data reflects a fixed sensor placement or accounts for variability in measurement location (e.g., position on the chest wall, distance from major airways, contact pressure). The changes in location and contact pressure would also alter the effective geometry of the acoustic cavity, which could in turn affect the acoustic transfer function. Furthermore, since respiratory sound intensity and spectral content are known to vary substantially with auscultation site, please clarify the placement protocol used during data collection and discuss how sensitivity to placement

variability might affect the robustness and generalizability of the system in practical, unsupervised use.

2. Performance of ML classification model. The 5-class confusion matrix (Fig. 4B) shows strong accuracy for the genuine respiratory-sound categories, but a markedly lower accuracy (65%) for the "non-breathing" class, with substantial bidirectional confusion. The true non-breathing recordings are frequently misclassified as crackle (12%), wheezing (9%), or healthy (9%), while other classes are misclassified as non-breathing in cases (7, 7, 5, 1%). Since the non-breathing class likely functions as a reject state that suppresses inappropriate actuation, this level of confusion has direct implications for false-positive/ false-negative alert rates that the manuscript should quantify explicitly. It is also not clear what acoustic conditions constitute the non-breathing class (e.g., speech, ambient noise, motion artifact, breath-holding)

3. Therapeutic benefit of vibrotactile feedback. The design of the alert actuation, which maps wheezing, crackles, and rhonchi to distinct vibration profiles, is an interesting and intuitive approach for communicating respiratory events to the user. However, the manuscript would benefit from a clearer discussion of the intended clinical or therapeutic rationale for this feedback design beyond simply notifying the user of the detected respiratory state. For example, do the authors envision that the mimetic vibration patterns may enhance patient awareness, facilitate faster behavioral responses, or provide other practical advantages compared with arbitrary haptic patterns or a simple on/off alarm? If these potential benefits have not yet been experimentally evaluated, it would be helpful to explicitly present them as motivating hypotheses or future directions rather than established capabilities.

4. Human-subject validation for the personalized guidance. The personalized feedback mode—reproducing an individual's resting cardiorespiratory rhythm as a tactile "reference rhythm" during stress—is one of the most compelling aspects of the manuscript. As currently presented, however, its potential relaxation or therapeutic benefit is primarily supported by prior literature (Lines 436–437) rather than by data obtained using the proposed system. Although a dedicated human-subject study would further strengthen this aspect of the work, it may also be sufficient for the authors to more clearly distinguish between the demonstrated functionality of the device and the anticipated therapeutic benefits of the personalized guidance. Clarifying this distinction and appropriately moderating the corresponding claims, if necessary, would improve the presentation of the work.

Minor Comments

1. Please provide detailed specifications of the MEMS microphone used in the device.

2. Could you clarify the rationale for choosing 700Hz for acoustic cavity simulation in Fig. S2, and could you explain the basis on which n=1.5 was selected as the optimal value in this simulation?

3. Could you clarify the end-to-end latency of the sensing-classification-actuation loop?

4. Consider adding a discussion for long-term/home use relevant to chronic-disease management, such as skin contact stability, and battery life